

ADITHYA SIVAKUMAR

Mechanical Design Engineer | CAE & FEA | CATIA V5 & SolidWorks

📍 Gothenburg, Sweden ✉ adithyasvk@outlook.com ☎ +46 767 485 838 📁 Portfolio 🌐 in LinkedIn



SUMMARY

Mechanical Engineer with 3+ years' experience in **design, analysis, and product development** of mechanical systems from concept to production. Skilled in CAD (CATIA V5, SolidWorks), CAE modelling and simulation, and cross-functional product development. Experienced in feasibility studies, **industrial design principles (DFM/DFA)**, and **compliance with industry standards**. Proactive, solution-oriented engineer with strong analytical, problem-solving, communication, and project management skills.

TECHNICAL COMPETENCE *(Self estimation 1-5)*

Design tools: **5** – CATIA V5, SolidWorks, Siemens NX ; **4** – Autodesk Inventor, Autodesk Fusion 360 ; **3** – AutoCAD, RapidHarness

Simulation tools: **5** – ANSA, ANSYS Mechanical, Maxwell, ElmerFEM ; **4** – COMSOL Multiphysics, Simulink ; **3** – ANSYS Fluent, Gmsh

Programming language: **5** – MATLAB; **4** – Python ; **Other tools:** **5** – Teamcenter PLM, MS Office, Prusa Slicer, CANalyzer

EDUCATION

Master of Science: Mobility Engineering (Specialization in Mechanical Engineering)

Aug 2023 – Oct 2025

Chalmers University of Technology - Gothenburg, Sweden

Bachelor of Technology: Mechanical Engineering (CGPA - 8.98/10)

Jul 2017 – Jul 2021

Amrita Vishwa Vidyapeetham - Coimbatore, India

EXPERIENCE

Internship - Research Assistant in Computational Engineering

Jönköping, Sweden

Jönköping University - Department of Product Development, Production and Design

Mar 2026 – Present

- Implementing Digital Image Correlation in Python for **real-time stress-strain curve estimation** in Universal Testing Machine.
- Mechanical design for a robust and adjustable camera mounting for DIC testing including CAD, prototyping and testing.

Summer Internship – Battery Hardware design

Gothenburg, Sweden

Volvo Cars (Battery Electrical Distribution and Battery Concept team)

June 2024 – Aug 2024

- Developed concept and mechanical design for a universal test rig for electrical and thermal validation of High-Voltage busbars.
- Benchmarked** competitor battery packs across structure, cooling, safety, electronics, and documentation to support design.

Project Engineer - Powertrain and Low Voltage

Gothenburg, Sweden

Chalmers Formula Student [\(CFS\)](#)

Sept 2023 – Sept 2024

- Head of department for **Powertrain** and Low Voltage systems (LV) subgroup; represented in **weekly design reviews**.
- Design of plastics and sheet metal**; enclosures, brackets, and cooling for LV electronics; 3D printing for rapid prototyping.
- Performed **mechanical integration of electronics** in enclosures with EMI, clearances, thermal, manufacturing and assembly.
- Conducted **FEM structural analysis** on mountings using strength calculations, FEA, and topology optimization using ANSYS.
- Achieved **18% weight savings** in mounting, **25% PC heat sink temperature drop** and developed robust **IP65 enclosures**.

R&D Engineer - Electric Machines

Chennai, India

Industrial Consultancy and Sponsored Research, Indian Institute of Technology-Madras

Jul 2021 – May 2023

- Project lead** for 2 kW Ferrite-Assisted PMSM motor for power drills; full product development from concept to production
- Prepared 3D CAD models and 2D drawings with **GD&T** and **Dfm/Dfa**. Selected suitable materials and manufacturing methods.
- Conducted **static, dynamic and fatigue FEA** of machine elements for performance, efficiency, reliability and durability.
- Prepared technical documentation including **BOM, specifications, DFMEA, and ECR/ECN**. Collaborated with cross-functional teams; suppliers and manufacturing teams for **prototype testing and validation**, troubleshooting, and **feasibility studies**.
- Led project management; ensure **compliance with industry standards and product improvements** within timelines.
- Achieved 5% weight reduction, 2% cost reduction** and 10% stress drop; improving FOS and product competitiveness.

ACADEMIC PROJECTS

Master thesis - Realistic Modelling of Li-ion Batteries for Calculating High Frequency Impedance

Gothenburg, Sweden

Volvo Cars (Virtual Methods & Analysis team) [Thesis Report](#)

Jan 2025 – Sept 2025

- Developed a physics-based high-fidelity 3D FEM model to characterize the **multiphysics behavior** of a Li-ion cell across **electric, magnetic, structural mechanics and thermal** domains in ElmerFEM using HPC in Linux. Mesh generated in ANSA.

- Developed an AI-based **Multi-Objective Genetic Algorithm automation workflow in Python** to optimize cell design.
- Identified regions of peak electric, magnetic, and displacement fields, along with von Mises stress and temperature contours.
- **Validated simulation solver** setup against experimental data at high frequency with less than 11% deviation in impedance.

Powertrain In-charge (Mechanical)

Coimbatore, India

Student Chapter, Society of Automobile Engineers - Range Extended Electric Vehicle ([SAE-REEV](#))

Dec 2018 – Feb 2020

- Objective was to design, fabricate and compete with a Series - Hybrid Electric Vehicle (HEV). Comprised a 20-day Internship at SAEINDIA. Led the mechanical powertrain subgroup in **powertrain sizing, design, procurement, assembly, and testing**.
- Performed CAD and FEA on differential sprocket and powertrain mounts. Performed IC Engine calibration and testing.
- Prepared Design Failure Mode and Effects Analysis (DFMEA), ECO/ECN, BOM, **root cause analyses and troubleshooting**.

Publications

Effect of passive cooling techniques on Energy Performance and Occupant Comfort for a Building in Warm-Humid Climate, Springer Nature, Singapore, [10.1007/978-981-19-3467-4_18](#)

Sept 2022

S. Karthick, **Adithya Sivakumar**, S. Bhanu Chander, S. Bharathwaj, B. V. V. Nagendra Kumar

Authored and presented scientific article at international conference. Applied advanced CFD simulations to evaluate thermal comfort and energy performance as part of peer-reviewed published research.

ACHIEVEMENTS

- Secured position P2 and P3 in the Driverless category at FS Germany and FS East competitions respectively in 2024.
- Received Best Paper Award for presenting the publication at International Conference on Energy and Materials Technologies.

LANGUAGES

English - C2 level (Fluent), German - A2 level, Swedish - Beginner

REFERENCES

Available upon request